

2024, Mar. 2024 Release LOC:Acute Adult**Hematology/Oncology: Complications or Disease Progression**Utilization Benchmarks: *Multiple options available, see table below for options***Overview****Select Day**Episode Day 1 ⁽¹⁾Episode Day 2 ⁽⁸⁶⁾Episode Day 3-X ⁽¹⁰⁸⁾**Discharge Screens** ⁽¹¹³⁾**Utilization Benchmarks****Notes**

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Overview**Instructional:**

This subset is intended to be used to complete a review of a patient with a hematologic and/or oncologic disease who is experiencing complication(s) related to outpatient treatment or progression of their disease. This may include the following complication(s):

- Brain malignancy or metastasis and associated neurological manifestation(s)
- Cerebral edema
- Cytokine release syndrome
- Graft versus host disease (GVHD)
- Hyperviscosity syndrome
- Jaundice or hyperbilirubinemia
- Malignant spinal cord compression (MSSC)
- Neurotoxicity
- Neutropenia
- Severe mucositis
- Thrombocytopenia
- Tumor lysis syndrome

Patients with malignancies admitted for chemotherapy, radiation, or immunotherapy (including CAR T-cell therapy) can be reviewed in the Hematology/Oncology: Treatments subset. Patients receiving inpatient chemotherapy, radiation, bone marrow or stem cell transplants, or immunotherapies can be reviewed in the Hematology/Oncology: Treatments subset. Patients with dehydration or gastroenteritis can be reviewed in the Dehydration or Gastroenteritis subset. Patients admitted for hypercalcemia can be reviewed in the Electrolyte and Mineral Imbalance subset. Patients admitted for actual or suspected Hemolytic Uremic Syndrome can be reviewed in the Hematology/Oncology: Hemolytic Uremic Syndrome subset.

Informational:

Brain and spinal malignancies are cancerous tumors that grow rapidly, invading adjacent normal structures, causing mass effects resulting in a single or combination of neurological impairments. They can be either primary or secondary. Primary tumors originate within the CNS structure, such as the brain or spinal cord. Secondary tumors arise from malignancies located elsewhere in the body that metastasizes to the brain or spinal cord via the bloodstream. Commonly, these include primary malignancies from the breast, lung, melanoma, or colon. Cytokine release syndrome (CRS) is an acute systemic inflammatory syndrome usually characterized by fever and

multiorgan dysfunction and most commonly seen in those patients who have received CAR T-cell therapy as an adverse effect. CRS can be life-threatening; the onset usually occurs within one day to 14 days after CAR T-cell infusion.

Symptoms could last up to 10 days, and monitoring and treatment are required. It is more common with patients with certain risk factors, including higher tumor burden, active infection at the time of CAR T-cell infusion, or type of CAR T-cell therapy utilized. Different grades of CRS have been developed based on reaction and symptomology.

Tumor lysis syndrome (TLS) occurs when rapid lysis of malignant cells into the bloodstream leads to significant metabolic and deleterious abnormalities. TLS typically occurs 12-72 hours after the initiation of chemotherapy. This syndrome is frequently associated with hematologic malignancies such as leukemia but may occur in the presence of some solid tumors. Prevention and aggressive management of diagnosed cases are necessary to avoid severe complications (e.g., cardiac arrhythmias, seizures, renal failure, and death). Some of the risk factors include advanced age and chronic renal insufficiency.

Neutropenia occurs when there is an abnormally low number of neutrophils in the bloodstream, defined as an absolute neutrophil count (ANC) starting at < 1500/cu.mm with a severe threshold starting at under 500/cu.mm. Neutropenia usually occurs after chemotherapy and radiation, where it causes agranulocytosis, putting patients at increased risk of developing infections from opportunistic organisms.

Thrombocytopenia is a platelet count of less than 150,000/cu.mm. There are many causes of low platelet counts. However, this subset captures thrombocytopenia caused by chemotherapy and complications from certain malignancies such as leukemia. This syndrome can lead to bleeding, and blood transfusion or corticosteroids may be required.

Oral mucositis is a common side effect of cancer treatments. Over one-third of cancer patients experience moderate or severe mucositis during cancer treatment. If untreated, mucositis can lead to anorexia, weight loss, malnutrition, and dehydration.

Graft versus Host Disease (GVHD) is a transplant complication that occurs when immunocompetent cells from a non-identical donor are grafted into an immunosuppressed recipient, and the immunocompetent donor/graft cells recognize alloantigens in the host cells leading to Graft vs. Host (GVH) reaction. Classic acute GVHD typically presents during the first 100 days after a hematopoietic stem cell transplantation (HSCT). Classic chronic GVHD presents beyond the first 100 days post-HSCT and may include overlap syndrome with acute GVHD features. Classic acute GVHD affects the skin, gut, and liver. Diagnosis is strongly supported if symptoms include exanthema (an eruptive skin rash oftentimes associated with a fever or systemic symptoms), diarrhea, and elevated bilirubin levels.

Evaluation and Treatment:

Symptoms of a brain or spinal tumor can differ depending on the region involved. Symptoms can include headache, seizures, visual deficits, ataxia, motor weakness, changes in speech, sensory abnormalities, nausea or vomiting, and mental status changes. Imaging modalities for brain and spinal tumors include computed tomography (CT) or magnetic resonance imaging (MRI). For a definitive diagnosis, a biopsy is recommended. Imaging will provide evidence of cerebral edema as a complication of the tumor mass. Treatment for malignant tumors and complications may include radiation, corticosteroids, chemotherapy, and surgery.

Cytokine release syndrome (CRS) symptoms can vary from fever, generalized malaise, and fatigue to hypotension, where fluids and/or vasopressors, hypoxemia, and respiratory failure may be required at higher levels of care. Treatment for CRS may include corticosteroids and/or IL-6 antagonist infusion.

Suspicion of Tumor lysis syndrome (TLS) may require checking labs such as uric acid, potassium, phosphorus, and calcium. Other manifestations required for diagnosing TLS include organ failure from renal impairment (most common), cardiac arrhythmias, and/or seizure(s) that occur after cancer-related therapies. Symptoms include nausea, vomiting, heart palpitations, edema, fatigue, and neuromuscular irritability. Treatment of tumor lysis syndrome is targeted at the specific metabolic disorder and may include dialysis.

For neutropenia and thrombocytopenia, importance should be given to alleviating infection risk and ruling out other potential causes of thrombocytopenia, such as sepsis. Platelet transfusion may be warranted, and in cases of disseminated intravascular coagulation with bleeding, fresh frozen plasma and/or cryoprecipitate transfusion could be required. Corticosteroids may be needed where thrombocytopenia manifests as a complication related to certain malignancies or chemotherapeutics.

Mucositis can manifest in varied degrees, with the main symptomology consisting of nausea, vomiting, and severe oropharyngeal pain. A patient may develop severe dehydration and weight loss as a complication due to the inability to tolerate oral intake. Neutropenia also plays a role in the development of mucositis.

References: (Ajani et al., J Natl Compr Canc Netw 2022, 20: 167-92; Pollyea et al., Journal of the National Comprehensive Cancer Network 2021, 19: 16-27; Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽²⁾● **Orthostatic hypotension, All:** ^(3, 4)

- Sustained drop in blood pressure within 3 min of sitting or standing
- Systolic BP drop ≥ 20 mmHg or diastolic BP drop ≥ 10 mmHg
- IV fluid bolus ⁽⁵⁾

● **Vomiting, persistent and unresponsive to ≥ 2 doses of antiemetic and, Both:** ⁽⁶⁾● **Medication, $\geq$ One:**

- Antiemetic $\geq 3x/24h$
- Dexamethasone ⁽⁷⁾
- Serotonin antagonist at least daily
- IV fluid ^(5, 8)

● **Acute pain and failed outpatient analgesic and, One:** ^(6, 9)

- Analgesic 2-3x/24h ⁽¹⁰⁾

● **ACUTE, $\geq$ One:** ⁽¹¹⁾

- Requiring surgical intervention (use **General Surgical subset**)

● **AKI and, Both:** ⁽¹²⁾● **Finding, $\geq$ One:**

- Urine output < 0.5 mL/kg/h

● **Creatinine, One:**

- $\geq 1.5x$ baseline and chronic kidney disease
- $\geq 1.5x$ ULN ⁽¹³⁾

- GFR $> 25\%$ (0.25) decrease from baseline ⁽¹⁴⁾

- Dialysis initiated

● **Intervention, $\geq$ One:**

- Volume expander $\leq 2d$ ^(15, 16)

- Diuretic $\geq 2x/24h$, $\leq 3d$

● **Medication adjustment or discontinuation $\leq 3d$ and, $\geq$ One:** ⁽¹⁷⁾

- Diuretic (includes PO)

- Nephrotoxic agent (includes PO) ⁽¹⁸⁾

- Dialysis (initial) and $\leq 7d$ since initiation ⁽¹⁹⁾

- Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽²⁰⁾

● **Brain malignancy or metastasis, actual or suspected and, Both:**● **Finding, $\geq$ One:**

- Ataxia
- Headache
- Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ^(21, 22)
- Paralysis ⁽²³⁾
- Paresis ⁽²³⁾
- Seizure

● **Intervention, $\geq$ One:**

- Anticonvulsant

- Corticosteroid

● **Dyspnea and, One:**

- O₂ sat $\leq 89\%$ (0.89) and $<$ baseline requiring supplemental oxygen ⁽¹⁴⁾

● **Fever, new onset and, Both:** ⁽²⁴⁾

- Temperature $\geq 100.4^\circ\text{F}$ (38.0°C) ⁽²⁵⁾

● **Intervention, $\geq$ One:**

- Culture pending $\leq 2d$

- Drug-induced fever suspected and precipitating drug discontinued $\leq 3d$ ⁽²⁶⁾
- Imaging study $\leq 24h$
- GVHD (Grade I or II) and, **Both:** ⁽²⁷⁾
 - Immunosuppressant ⁽²⁸⁾
- Intervention, $\geq$ **One:**
 - Medication adjustment (includes PO) $\leq 3d$ ^(17, 29)
 - New medication initiation ⁽³⁰⁾
- Hyperviscosity syndrome and, **Both:** ⁽³¹⁾
 - Finding, $\geq$ **One:**
 - Mucosal bleeding
 - Neurological symptoms
 - Visual changes
 - Intervention, **All:**
 - Apheresis ⁽³²⁾
 - Hydration or nutrition, $\geq$ **One:**
 - IV fluid ^(5, 8)
 - Advancing diet as tolerated
 - Neurological assessment at least 3x/24h ⁽³³⁾
- Intractable diarrhea or colitis and, $\geq$ **One:** ^(34, 35)
 - Corticosteroid ⁽³⁶⁾
 - Immunosuppressant therapy (includes PO and SC) ⁽³⁷⁾
 - IV fluid bolus ⁽⁵⁾
- Intractable pain and analgesic and, $\geq$ **One:** ^(10, 38)
 - $\geq 4x/24h$
 - Continuous
- Jaundice or hyperbilirubinemia and, **Both:** ⁽³⁹⁾
 - Finding, $\geq$ **One:**
 - PT $\geq 1.5x$ ULN
 - INR ≥ 2.0
 - Total bilirubin $\geq 2x$ ULN
 - Intervention, $\geq$ **One:**
 - ERCP scheduled or performed within last 24h
 - Fresh frozen plasma (FFP)
 - Liver biopsy
 - MRCP scheduled or performed within last 24h
 - Percutaneous drainage performed within last 24h
- Malignant pleural effusion and, **Both:** ⁽⁴⁰⁾
 - O₂ sat 89-91%(0.89-0.91)
 - Intervention, $\geq$ **One:**
 - Chest tube insertion ^(41, 42, 43)
 - Chest tube and sclerosing agent ^(41, 44, 45)
- Neutropenia, **All:** ^(46, 47)
 - ANC, **One:** ⁽⁴⁸⁾
 - $< 500/cu.mm(500 \times 10^6/L)$
 - $< 1000/cu.mm(1000 \times 10^6/L)$ with documented decline ⁽⁴⁹⁾
 - Anti-infective
 - Temperature $\geq 100.4^\circ F(38.0^\circ C)$ ⁽²⁵⁾
- Severe mucositis and inability to take PO and, $\geq$ **One:** ⁽⁵⁰⁾
 - IV fluid ^(5, 8)
 - Parenteral nutrition
 - Tube feeding
- Spinal cord compression by imaging requiring corticosteroid ^(51, 52)

- Thrombocytopenia (excludes known Von Willebrand disease) and, **Both:** ^(53, 54)
 - Finding, **≥ One:**
 - Platelet count < 20,000/cu.mm($20 \times 10^9/L$)
 - Purpura or petechiae, progressive
 - Intervention, **≥ One:**
 - Blood product transfusion
 - Corticosteroid
 - Immunoglobulin ⁽⁵⁵⁾
- Tumor lysis syndrome and, **Both:** ⁽⁵⁶⁾
 - Finding, **≥ Two:**
 - Calcium < 7.0 mg/dL(1.8 mmol/L)
 - Potassium > 6.0 mg/L(6.0 mmol/L) and no ECG changes
 - Phosphate > 4.5 mg/dL(1.5 mmol/L)
 - Uric acid > 8.0 mg/dL(475.9 $\mu\text{mol/L}$)
 - Intervention, **One:**
 - Dialysis not required, **Both:**
 - IV fluid ^(8, 57)
 - Medication, **One:** ⁽⁵⁸⁾
 - Allopurinol (includes PO)
 - Rasburicase
 - Hemodialysis $\leq 7d$ since initiation
- INTERMEDIATE, **≥ One:** ⁽⁵⁹⁾
 - Cytokine release syndrome, actual or suspected and, **Both:** ^(60, 61)
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ and, **≥ One:** ⁽²⁵⁾
 - O_2 sat $\leq 89\%(0.89)$ and < baseline requiring supplemental oxygen ⁽¹⁴⁾
 - Systolic BP ≥ 90 mmHg
 - Intervention, **≥ One:**
 - Anticytokine therapy
 - Corticosteroid (includes PO)
 - IV fluid ^(5, 8)
 - GVHD and, **One:** ⁽²⁷⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽⁶²⁾
 - Grade III and immunosuppressant ⁽²⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(63, 64)
 - Neurotoxicity, actual or suspected and, **Both:**
 - Finding, **≥ One:**
 - Aphasia ⁽⁶⁵⁾
 - Apraxia
 - Difficulty counting numbers backward
 - Disorientation to time or place
 - New or worsening tremor
 - Unable to write a sentence
 - Neurological assessment every 3-4h and, **≥ One:**
 - IV fluid ^(5, 8)
 - Corticosteroid (includes PO)
 - NIPPV ⁽⁶⁶⁾
 - Oxygen $\geq 40\%(0.40) \leq 2d$ ⁽⁶⁷⁾
 - Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant, continuous
 - Tumor lysis syndrome and, **≥ One:** ⁽⁵⁶⁾
 - Arrhythmia and, **≥ One:**

- IV antiarrhythmic and hemodynamically stable
- Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk ≤ 24 h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3 d ⁽⁶⁸⁾
- Hyperkalemia, **All:** ^(69, 70)
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
- ECG changes, $\geq$ **One:** ⁽⁷⁰⁾
 - Peaked T waves
 - PVCs > 6 /min
- Intervention, $\geq$ **One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷¹⁾
- **CRITICAL**, $\geq$ **One:** ⁽⁷²⁾
 - Anticonvulsant and, $\geq$ **One:**
 - Recurrent seizure ⁽⁷³⁾
 - Refractory seizure ⁽⁷³⁾
 - Arrhythmia, $\geq$ **One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, $\geq$ **One:** ⁽⁷⁴⁾
 - Chest pain
 - Heart failure and, $\geq$ **One:**
 - Arterial $P_{O_2} < 56$ mmHg(7.4 kPa)
 - O_2 sat $< 89\%$ (0.89) and $<$ baseline
 - Hemodynamic instability, $\geq$ **One:**
 - Heart rate > 120 /min, sustained ⁽⁷⁵⁾
 - Systolic blood pressure, $\geq$ **One:**
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽¹⁴⁾
 - Labile ⁽⁷⁶⁾
 - MAP < 65 mmHg
- Cerebral edema and, $\geq$ **One:**
 - Corticosteroid
- Hyperosmolar therapy, $\geq$ **One:** ⁽⁷⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%$ (0.03) solution
- ICP monitoring (includes OR placement)
- Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁾
- CRRT ⁽⁷⁸⁾
- DIC and, **All:** ⁽⁷⁹⁾
 - Fibrinogen degradation product (FDP) increasing
 - Fibrinogen decreasing
 - Blood product transfusion
- GVHD, Grade IV and, $\geq$ **One:** ⁽²⁷⁾
 - Apheresis ⁽³²⁾
 - Immunosuppressant ⁽²⁸⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽⁸⁰⁾
 - MAP < 65 mmHg

- Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽⁸¹⁾
- HFNC and respiratory monitoring every 1-2h ^(63, 64, 67)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽⁶⁶⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One**:
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Tumor lysis syndrome and, **Both**: ⁽⁵⁶⁾
 - Finding, ≥ **One**:
 - Calcium < 5.0 mg/dL(1.25 mmol/L) ^(82, 83)
 - Hyperkalemia, **Both**:
 - Potassium > 6.0 mEq/L(6.0 mmol/L)
 - Finding, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁸⁴⁾
 - Widening QRS
 - Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One**:
 - Calcium repletion
 - IV fluid resuscitation ≤ 2d ⁽⁸⁵⁾
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷¹⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One**: ⁽⁸⁶⁾
 - **OBSERVATION, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽⁸⁷⁾
 - Pain controlled or manageable ⁽⁸⁸⁾
 - Tolerating PO or nutritional route established ⁽⁸⁹⁾
 - Vomiting resolved
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Orthostatic hypotension, **All**: ^(3, 4)
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic BP drop ≥ 20 mmHg or diastolic BP drop ≥ 10 mmHg
 - IV fluid bolus ⁽⁵⁾
 - Vomiting, persistent and, **Both**: ⁽⁶⁾
 - Medication, ≥ **One**:
 - Antiemetic ≥ 3x/24h
 - Dexamethasone ⁽⁷⁾
 - Serotonin antagonist at least daily
 - IV fluid ^(5, 8)
 - Acute pain and analgesic 2-3x/24h ^(9, 10)
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽⁸⁷⁾
 - Afebrile

- Anti-infective, **One:**
 - Arranged
 - Completed
 - Not indicated
- Medication regimen established and tolerated
- Mental status or GCS within normal limits or returned to baseline ⁽²²⁾
- Lab values within acceptable limits ⁽⁹⁰⁾
- Tolerating PO or nutritional route established ^(89, 91)
- Vital signs stable or within acceptable limits ⁽⁹⁰⁾
- Complication or comorbidity, **One:** ⁽⁹²⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁹²⁾
- Complication or comorbidity and clinically stable for discharge, **≥ One:**
 - AKI and, **≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁹⁰⁾
 - Outpatient dialysis regimen established
 - Decrease in frequency and severity of diarrhea
 - Dyspnea resolved and, **≥ One:**
 - O₂ sat within acceptable limits ⁽⁹⁰⁾
 - Home oxygen arranged
 - Jaundice improved or stabilized
 - Malignant pleural effusion resolving and, **One:**
 - Chest tube to Heimlich valve
 - Chest tube removed and no reaccumulation of air or fluid by CXR
 - Pain controlled or manageable ⁽⁸⁸⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - Requiring surgical intervention (**use General Surgical subset**)
- AKI and, **Both:** ⁽¹²⁾
 - Finding, **≥ One:**
 - Urine output < 0.5 mL/kg/h
 - Creatinine, **One:**
 - $\geq 1.5\times$ baseline and chronic kidney disease
 - $\geq 1.5\times$ ULN ⁽¹³⁾
 - GFR > 25%(0.25) decrease from baseline ⁽¹⁴⁾
 - Dialysis initiated
 - Intervention, **≥ One:**
 - Volume expander $\leq 2d$ ^(15, 16)
 - Diuretic $\geq 2x/24h$, $\leq 3d$
 - Medication adjustment or discontinuation $\leq 3d$ and, **≥ One:** ⁽¹⁷⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽¹⁸⁾
 - Dialysis (initial) and $\leq 7d$ since initiation ⁽¹⁹⁾
 - Dialysis discontinued requiring lab monitoring $\leq 2d$ since discontinuation ⁽²⁰⁾
- Blood glucose $\leq 24h$, **All:** ⁽⁹³⁾
 - Finding, **≥ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(93, 94)
- Brain malignancy, brain metastasis or spinal cord compression by imaging and, **≥ One:**
 - Anticonvulsant
 - Corticosteroid $\leq 2d$ since initiation
- COPD and, **All:**

- Finding, **≥ One:**
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁴⁾
 - Arterial Po₂ ≤ 60 mmHg(8.0 kPa) ⁽⁹⁵⁾
 - Arterial or venous Pco₂ > baseline ^(14, 95)
 - Bronchodilator ≥ 4x/24h ⁽⁹⁶⁾
 - Corticosteroid (includes PO or inhaled)
- Oxygenation, **One:**
 - O₂ sat ≥ 92%(0.92) or baseline ⁽¹⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(14, 67)
- Dyspnea and, **One:**
 - O₂ sat ≤ 89%(0.89) and < baseline requiring supplemental oxygen ⁽¹⁴⁾
- Fever, new onset and, **Both:** ⁽²⁴⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽²⁵⁾
- Intervention, **≥ One:**
 - Culture pending ≤ 2d
 - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d ⁽²⁶⁾
 - Imaging study ≤ 24h
- GVHD and immunosuppressant and, **≥ One:** ^(27, 28, 97)
 - Conversion from IV to PO ≤ 2d ⁽⁹⁸⁾
 - Medication adjustment (includes PO) ≤ 3d ^(17, 29)
 - New medication initiation ⁽³⁰⁾
- Heart failure and, **Both:** ⁽⁹⁹⁾
 - Dyspnea > baseline and, **≥ One:** ⁽¹⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁴⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽¹⁰⁰⁾
 - Intervention, **≥ One:** ⁽¹⁰¹⁾
 - Diuretic, **One:** ⁽¹⁰²⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, **≥ One:** ⁽¹⁷⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹⁴⁾
 - Creatinine increased from baseline ⁽¹⁴⁾
 - Dialysis
- Hyperviscosity syndrome and, **All:** ⁽³¹⁾
 - Apheresis ⁽³²⁾
- Hydration or nutrition, **≥ One:**
 - IV fluid ^(5, 8)
 - Advancing diet as tolerated
 - Neurological assessment at least 3x/24h, ≤ 2d ⁽³³⁾
- Inadequate oral intake ≤ 7d and, **≥ One:** ⁽¹⁰³⁾
 - Antiemetic, **≥ One:**
 - Antiemetic ≥ 3x/24h
 - Dexamethasone ⁽⁷⁾
 - Serotonin antagonist at least daily
 - IV fluid ^(5, 8)
 - Parenteral nutrition
 - Tube feeding
- Intractable diarrhea or colitis and, **≥ One:** ^(34, 35)
 - Corticosteroid ⁽³⁶⁾
 - Immunosuppressant therapy (includes PO and SC) ⁽³⁷⁾
 - IV fluid bolus ⁽⁵⁾
- Intractable pain and, **≥ One:**

- Conversion from IV to PO or transdermal $\leq 24\text{h}$
- Analgesic $\leq 2\text{d}$, $\geq$ **One:** ⁽¹⁰⁾
 - $\geq 4\text{x}/24\text{h}$
 - Continuous
- Jaundice or hyperbilirubinemia and, **Both:**
 - Finding, $\geq$ **One:**
 - $\text{INR} \geq 2.0$
 - $\text{PT} \geq 1.5\text{x ULN}$
 - Total bilirubin $\geq 2\text{x ULN}$
 - Intervention, $\geq$ **One:**
 - ERCP scheduled or performed within last 24h
 - Fresh frozen plasma (FFP)
 - Liver biopsy
 - MRCP scheduled or performed within last 24h
 - Percutaneous drainage performed within last 24h
- Malignant pleural effusion and chest tube and, $\geq$ **One:** ^(40, 44)
 - Drainage $\geq 100\text{ mL/d}$
 - Requiring repositioning within last 24h
 - Sclerosing agent or chemotherapy ⁽⁴⁵⁾
 - Suction, continuous
 - Water seal within last 24h
- Neutropenia, **All:** ⁽⁴⁶⁾
 - ANC, **One:** ⁽⁴⁸⁾
 - $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$
 - $< 1000/\text{cu.mm}(1000 \times 10^6/\text{L})$ with documented decline ⁽⁴⁹⁾
 - Anti-infective
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽²⁵⁾
- Pneumonia confirmed by imaging and, **Both:** ⁽¹⁰⁴⁾
 - Finding, $\geq$ **One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽²⁵⁾
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽²¹⁾
 - $\text{O}_2\text{ sat} \leq 91\%(0.91)$ and $<$ baseline ⁽¹⁴⁾
 - $\text{WBC} \geq 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$
 - $\text{WBC} \leq 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - $\text{ANC} < 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽⁴⁸⁾
 - Intervention, $\geq$ **One:** ⁽¹⁰⁵⁾
 - Requiring supplemental oxygen ⁽⁶⁷⁾
 - Anti-infective (includes PO)
 - Respiratory interventions at least $6\text{x}/24\text{h}$ ⁽¹⁰⁶⁾
- Thrombocytopenia and, $\geq$ **One:**
 - Blood product transfusion
 - Corticosteroid
 - Immunoglobulin ⁽⁵⁵⁾
- Tumor lysis syndrome and, $\geq$ **One:** ⁽⁵⁶⁾
 - Dialysis not required, **Both:**
 - IV fluid ^(8, 57)
 - Medication, **One:** ⁽⁵⁸⁾
 - Allopurinol (includes PO)
 - Rasburicase
 - Hemodialysis $\leq 7\text{d}$ since initiation
 - Post Critical care $\leq 24\text{h}$

- **INTERMEDIATE, ≥ One:**
 - Arrhythmia, ≥ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽⁶⁸⁾
 - Cytokine release syndrome and, ≥ **One:** ^(60, 61)
 - Anticytokine therapy
 - Corticosteroid (includes PO)
 - IV fluid ^(5, 8)
 - GVHD and, **One:** ⁽²⁷⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽⁶²⁾
 - Grade III and immunosuppressant ⁽²⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(63, 64)
 - Hyperkalemia, **All:** ^(69, 70)
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - ECG changes, ≥ **One:** ⁽⁷⁰⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷¹⁾
 - Neurological impairment or condition, **Both:** ⁽¹⁰⁷⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, ≤ 2d ⁽³³⁾
 - Neurotoxicity, actual or suspected and, **Both:**
 - Finding, ≥ **One:**
 - Aphasia ⁽⁶⁵⁾
 - Apraxia
 - Difficulty counting numbers backward
 - Disorientation to time or place
 - New or worsening tremor
 - Unable to write a sentence
 - Neurological assessment every 3-4h and, ≥ **One:**
 - IV fluid ^(5, 8)
 - Corticosteroid (includes PO)
 - NIPPV ⁽⁶⁶⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁶⁷⁾
 - Seizure and, **Both:**
 - Activity resolved
 - Anticonvulsant, continuous
- **CRITICAL, ≥ One:**
 - Anticonvulsant and, ≥ **One:**
 - Recurrent seizure ⁽⁷³⁾
 - Refractory seizure ⁽⁷³⁾
 - Arrhythmia, ≥ **One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, ≥ **One:** ⁽⁷⁴⁾
 - Chest pain

- Heart failure and, ≥ **One**:
 - Arterial $\text{Po}_2 < 56 \text{ mmHg}$ (7.4 kPa)
 - $\text{O}_2 \text{ sat} < 89\%$ (0.89) and $< \text{baseline}$
- Hemodynamic instability, ≥ **One**:
 - Heart rate $> 120/\text{min}$, sustained ⁽⁷⁵⁾
 - Systolic blood pressure, ≥ **One**:
 - $< 90 \text{ mmHg}$
 - $< 110 \text{ mmHg}$ with chronic hypertension
 - $> 30 \text{ mmHg}$ decrease from baseline ⁽¹⁴⁾
 - Labile ⁽⁷⁶⁾
 - MAP $< 65 \text{ mmHg}$
- Cerebral edema and, ≥ **One**:
 - Corticosteroid
 - Hyperosmolar therapy, ≥ **One**: ⁽⁷⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%$ (0.03) solution
 - ICP monitoring (includes OR placement)
- Coma, stupor, obtundation, or GCS ≤ 8 ⁽²²⁾
- CRRT ⁽⁷⁸⁾
- DIC and blood product transfusion ⁽⁷⁹⁾
- GVHD, Grade IV and, ≥ **One**: ⁽²⁷⁾
 - Apheresis ⁽³²⁾
 - Immunosuppressant ⁽²⁸⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24\text{h}$ and, ≥ **One**: ⁽⁸⁰⁾
 - MAP $< 65 \text{ mmHg}$
 - Systolic blood pressure $< 90 \text{ mmHg}$ and $< \text{baseline}$
- Hemodynamic monitoring, invasive ⁽⁸¹⁾
- HFNC and respiratory monitoring every 1-2h ^(63, 64, 67)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽⁶⁶⁾
 - $\text{FiO}_2 > 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽¹⁰⁶⁾
- Tumor lysis syndrome and, **Both**: ⁽⁵⁶⁾
 - Finding, ≥ **One**:
 - Calcium $< 5.0 \text{ mg/dL}$ (1.25 mmol/L) ^(82, 83)
 - Hyperkalemia, **Both**:
 - Potassium $> 6.0 \text{ mEq/L}$ (6.0 mmol/L)
 - Finding, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁸⁴⁾
 - Widening QRS
 - Sodium, **One**:
 - $< 120 \text{ mEq/L}$ (120 mmol/L) and volume overload
 - $> 160 \text{ mEq/L}$ (160 mmol/L)
- Intervention, ≥ **One**:
 - Calcium repletion
 - IV fluid resuscitation $\leq 2\text{d}$ ⁽⁸⁵⁾
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis $\leq 7\text{d}$ since initiation

- Peritoneal dialysis $\leq$ 7d since initiation
- Glucose 50%(0.50) with insulin ⁽⁷¹⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3-X, One:** ⁽¹⁰⁸⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁸⁷⁾
 - Pain controlled or manageable ⁽⁸⁸⁾
 - Tolerating PO or nutritional route established ⁽⁸⁹⁾
 - Vomiting resolved
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For current condition use Episode Day 3 at a higher level of care
 - For a condition other than the current condition use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁸⁷⁾
 - Afebrile
 - **Anti-infective, One:**
 - Arranged
 - Completed
 - Not indicated
 - Medication regimen established and tolerated
 - Mental status or GCS within normal limits or returned to baseline ⁽²²⁾
 - Lab values within acceptable limits ⁽⁹⁰⁾
 - Tolerating PO or nutritional route established ^(89, 91)
 - Vital signs stable or within acceptable limits ⁽⁹⁰⁾
 - **Complication or comorbidity, One:** ⁽⁹²⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁹²⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - **AKI and, $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽⁹⁰⁾
 - Outpatient dialysis regimen established
 - Blood glucose within acceptable limits ⁽⁹⁰⁾
 - COPD exacerbation resolving
 - Decrease in frequency and severity of diarrhea
 - **Dyspnea resolved and, $\geq$ One:**
 - Home oxygen arranged
 - O₂ sat within acceptable limits ⁽⁹⁰⁾
 - **Functional impairment and, One:** ⁽¹⁰⁹⁾
 - **Discharge to home and, One:** ⁽¹¹⁰⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - GVHD resolving or stabilized
 - **Heart failure resolving and, Both:**
 - Dyspnea resolving or at baseline ⁽¹⁴⁾
 - O₂ sat $\geq$ 92%(0.92) or within acceptable limits ⁽⁹⁰⁾
 - **Hyperviscosity syndrome resolving and, Both:**
 - Neurological stability ⁽¹¹¹⁾
 - No evidence of bleeding
 - Jaundice improved or stabilized

- Malignant pleural effusion resolving and, **One:**
 - Chest tube to Heimlich valve
 - Chest tube removed and no reaccumulation of air or fluid by CXR
- No seizure activity or baseline ⁽¹⁴⁾
- Pain controlled or manageable ⁽⁸⁸⁾
- Spinal cord compression resolving and neurological stability ⁽¹¹¹⁾
- Tumor lysis syndrome resolved or stabilized
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Requiring surgical intervention (**use General Surgical subset**)
 - Stroke (**use Stroke subset**)
- AKI and, **Both:** ⁽¹²⁾
 - Finding, ≥ **One:**
 - Urine output < 0.5 mL/kg/h
 - Creatinine, **One:**
 - ≥ 1.5x baseline and chronic kidney disease
 - ≥ 1.5x ULN ⁽¹³⁾
 - GFR > 25%(0.25) decrease from baseline ⁽¹⁴⁾
 - Dialysis initiated
- Intervention, ≥ **One:**
 - Volume expander ≤ 2d ^(15, 16)
 - Diuretic ≥ 2x/24h, ≤ 3d
- Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** ⁽¹⁷⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽¹⁸⁾
 - Dialysis (initial) and ≤ 7d since initiation ⁽¹⁹⁾
 - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽²⁰⁾
- Blood glucose ≤ 24h, **All:** ⁽⁹³⁾
 - Finding, ≥ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(93, 94)
- Brain malignancy, brain metastasis or spinal cord compression by imaging and, ≥ **One:**
 - Anticonvulsant
 - Corticosteroid ≤ 2d since initiation
- Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ **One:** ^(109, 112)
 - Cognitive evaluation and retraining
 - OT evaluation and training
 - PT evaluation and training
 - Speech-language evaluation and training
 - Swallowing evaluation and retraining
- Seizure, new onset and, ≥ **One:**
 - Change in anti-convulsant medication or dose (includes PO) ≤ 2d
 - Initiation of anti-convulsant (includes PO) ≤ 2d
- COPD and, **All:**
 - Finding, ≥ **One:**
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁴⁾
 - Arterial Po₂ ≤ 60 mmHg(8.0 kPa) ⁽⁹⁵⁾
 - Arterial or venous Pco₂ > baseline ^(14, 95)
 - Bronchodilator ≥ 4x/24h ⁽⁹⁶⁾
 - Corticosteroid (includes PO or inhaled)
- Oxygenation, **One:**

- O₂ sat ≥ 92%(0.92) or baseline ⁽¹⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen ^(14, 67)
- Fever, new onset and, **Both:** ⁽²⁴⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽²⁵⁾
- Intervention, ≥ **One:**
 - Culture pending ≤ 2d
 - Drug-induced fever suspected and precipitating drug discontinued ≤ 3d ⁽²⁶⁾
 - Imaging study ≤ 24h
- GVHD and immunosuppressant and, ≥ **One:** ^(27, 28, 97)
 - Conversion from IV to PO ≤ 2d ⁽⁹⁸⁾
 - Medication adjustment (includes PO) ≤ 3d ^(17, 29)
 - New medication initiation ⁽³⁰⁾
- Heart failure and, **Both:** ⁽⁹⁹⁾
 - Dyspnea > baseline and, ≥ **One:** ⁽¹⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽¹⁴⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽¹⁰⁰⁾
 - Intervention, ≥ **One:** ⁽¹⁰¹⁾
 - Diuretic, **One:** ⁽¹⁰²⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** ⁽¹⁷⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽¹⁴⁾
 - Creatinine increased from baseline ⁽¹⁴⁾
 - Dialysis
- Hyperviscosity syndrome ≤ 6d since onset and, **Both:** ⁽³¹⁾
 - Finding, ≥ **One:**
 - Mucosal bleeding
 - Persistent neurological symptoms
 - Visual changes
 - Intervention, ≥ **One:**
 - Apheresis ⁽³²⁾
 - Hydration or nutrition, ≥ **One:**
 - IV fluid ^(5, 8)
 - Advancing diet as tolerated
 - Neurological assessment at least 3x/24h, ≤ 2d ⁽³³⁾
- Inadequate oral intake ≤ 7d and, ≥ **One:** ⁽¹⁰³⁾
 - Antiemetic, ≥ **One:**
 - ≥ 3x/24h
 - Dexamethasone ⁽⁷⁾
 - Serotonin antagonist at least daily
 - IV fluid ^(5, 8)
 - Parenteral nutrition
 - Tube feeding
- Intractable diarrhea or colitis and, ≥ **One:** ^(34, 35)
 - Corticosteroid ⁽³⁶⁾
 - Immunosuppressant therapy (includes PO and SC) ⁽³⁷⁾
 - IV fluid bolus ⁽⁵⁾
- Intractable pain and, ≥ **One:**
 - Conversion from IV to PO or transdermal ≤ 24h
 - Analgesic ≤ 2d, ≥ **One:** ⁽¹⁰⁾
 - ≥ 3x/24h
 - Continuous

- Jaundice or hyperbilirubinemia and, **≥ One:**
 - Fresh frozen plasma (FFP) and, **≥ One:**
 - PT $\geq 1.5\times$ ULN
 - INR ≥ 2.0
 - Total bilirubin $\geq 2\times$ ULN
 - Liver biopsy performed within last 24h
 - Percutaneous drain requiring repositioning within last 24h
 - Percutaneous drainage performed within last 24h
- Malignant pleural effusion and chest tube and, **≥ One:** ^(40, 44)
 - Drainage ≥ 100 mL/d
 - Requiring repositioning within last 24h
 - Sclerosing agent or chemotherapy ⁽⁴⁵⁾
 - Suction, continuous
 - Water seal within last 24h
- Neutropenia, **All:** ⁽⁴⁶⁾
 - ANC, **One:** ⁽⁴⁸⁾
 - $< 500/\text{cu.mm}(500\times 10^6/\text{L})$
 - $< 1000/\text{cu.mm}(1000\times 10^6/\text{L})$ with documented decline ⁽⁴⁹⁾
 - Anti-infective
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽²⁵⁾
- Pneumonia confirmed by imaging and, **Both:** ⁽¹⁰⁴⁾
 - Finding, **≥ One:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽²⁵⁾
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽²¹⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline ⁽¹⁴⁾
 - WBC $\geq 12,000/\text{cu.mm}(12\times 10^9/\text{L})$
 - WBC $\leq 4,000/\text{cu.mm}(4\times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - ANC $< 500/\text{cu.mm}(500\times 10^6/\text{L})$ ⁽⁴⁸⁾
 - Intervention, **≥ One:** ⁽¹⁰⁵⁾
 - Requiring supplemental oxygen ⁽⁶⁷⁾
 - Anti-infective (includes PO)
 - Respiratory interventions at least 6x/24h ⁽¹⁰⁶⁾
- Thrombocytopenia $\leq 7\text{d}$ since onset and, **Both:**
 - Finding, **≥ One:**
 - Bleeding actual or suspected and, **≥ One:**
 - Hct $< 25\%(0.25)$ or Hb < 8.3 g/dL(83 g/L)
 - Platelet count $< 60,000/\text{cu.mm}(60\times 10^9/\text{L})$
 - Heart rate 100-120/min, sustained ⁽⁷⁵⁾
 - Hct or Hb decreased last 24h
 - Orthostatic hypotension, **Both:** ⁽⁴⁾
 - Sustained drop in blood pressure within 3 min of sitting or standing ⁽⁷⁵⁾
 - Blood pressure, **≥ One:**
 - Systolic drop ≥ 20 mmHg
 - Diastolic drop ≥ 10 mmHg
 - PT $\geq 1.5\times$ ULN or INR ≥ 2.0
 - PTT $\geq 1.5\times$ ULN
 - Platelet count $< 50,000/\text{cu.mm}(50\times 10^9/\text{L})$
 - Purpura or petechiae, progressive or unknown etiology
 - Intervention, **≥ One:**
 - Blood product transfusion
 - Corticosteroid

- Immunoglobulin ⁽⁵⁵⁾
- Tumor lysis syndrome and, ≥ **One:** ⁽⁵⁶⁾
 - Dialysis not required, **Both:**
 - IV fluid ^(8, 57)
 - Medication, **One:** ⁽⁵⁸⁾
 - Allopurinol (includes PO)
 - Rasburicase
 - Hemodialysis ≤ 7d since initiation
 - Post Critical care ≤ 24h
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - Use **Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **One:** ⁽⁸⁷⁾
 - Cytokine release syndrome and, **All:**
 - Afebrile
 - Medication regimen established and tolerated
 - Mental status or GCS within normal limits or returned to baseline ⁽²²⁾
 - Lab values within acceptable limits ⁽⁹⁰⁾
 - Tolerating PO or nutritional route established ^(89, 91)
 - Vital signs stable or within acceptable limits ⁽⁹⁰⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Arrhythmia, ≥ **One:**
 - IV antiarrhythmic and hemodynamically stable
 - Antiarrhythmic conversion from IV to PO, **One:**
 - No proarrhythmic risk ≤ 24h
 - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d ⁽⁶⁸⁾
 - Cytokine release syndrome and, ≥ **One:** ^(60, 61)
 - Anticytokine therapy
 - Corticosteroid (includes PO)
 - IV fluid ^(5, 8)
 - GVHD and, **One:** ⁽²⁷⁾
 - Grade I or II and monoclonal antibody initiation ≤ 3 doses ⁽⁶²⁾
 - Grade III and immunosuppressant ⁽²⁸⁾
 - HFNC and respiratory monitoring every 3-4h ^(63, 64)
 - Hyperkalemia, **All:** ^(69, 70)
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - ECG changes, ≥ **One:** ⁽⁷⁰⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷¹⁾
 - Neurological impairment or condition, **Both:** ⁽¹⁰⁷⁾
 - Acute onset or deterioration
 - Neurological assessment every 3-4h, ≤ 2d ⁽³³⁾
 - Neurotoxicity, actual or suspected and, **Both:**
 - Finding, ≥ **One:**
 - Aphasia ⁽⁶⁵⁾
 - Apraxia

- Difficulty counting numbers backward
- Disorientation to time or place
- New or worsening tremor
- Unable to write a sentence
- **Neurological assessment every 3-4h and, ≥ One:**
 - IV fluid ^(5, 8)
 - Corticosteroid (includes PO)
- NIPPV ⁽⁶⁶⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽⁶⁷⁾
- **Seizure and, Both:**
 - Activity resolved
 - Anticonvulsant, continuous
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - Use Extended Stay subset
 - Refer for secondary review
- **CRITICAL, One:**
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - **Anticonvulsant and, ≥ One:**
 - Recurrent seizure ⁽⁷³⁾
 - Refractory seizure ⁽⁷³⁾
 - **Arrhythmia, ≥ One:**
 - Urgent electrical cardioversion performed within last 24h
 - Defibrillation performed within last 24h
 - Temporary pacemaker insertion performed within last 2d
 - **Symptomatic recurrent or refractory arrhythmia requiring antiarrhythmic and, ≥ One:** ⁽⁷⁴⁾
 - Chest pain
 - **Heart failure and, ≥ One:**
 - Arterial $PO_2 < 56$ mmHg(7.4 kPa)
 - O_2 sat $< 89\%(0.89)$ and $<$ baseline
 - **Hemodynamic instability, ≥ One:**
 - Heart rate $> 120/\text{min}$, sustained ⁽⁷⁵⁾
 - **Systolic blood pressure, ≥ One:**
 - < 90 mmHg
 - < 110 mmHg with chronic hypertension
 - > 30 mmHg decrease from baseline ⁽¹⁴⁾
 - Labile ⁽⁷⁶⁾
 - MAP < 65 mmHg
 - **Cerebral edema and, ≥ One:**
 - Corticosteroid
 - **Hyperosmolar therapy, ≥ One:** ⁽⁷⁷⁾
 - Mannitol
 - Hypertonic saline $\geq 3\%(0.03)$ solution
 - ICP monitoring (includes OR placement)
 - Coma, stupor, obtundation or GCS ≤ 8 ⁽²²⁾
 - CRRT ⁽⁷⁸⁾
 - DIC and blood product transfusion $\leq 7d$ since onset ⁽⁷⁹⁾
 - **GVHD, Grade IV, $\leq 7d$ since onset and, ≥ One:** ⁽²⁷⁾
 - Apheresis ⁽³²⁾
 - Immunosuppressant ⁽²⁸⁾
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$ and, ≥ One:** ⁽⁸⁰⁾

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽⁸¹⁾
- HFNC and respiratory monitoring every 1-2h ^(63, 64, 67)
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽⁶⁶⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁰⁶⁾
- Tumor lysis syndrome and, **Both:** ⁽⁵⁶⁾
 - Finding, ≥ **One:**
 - Calcium < 5.0 mg/dL(1.25 mmol/L) ^(82, 83)
 - Hyperkalemia, **Both:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L)
 - Finding, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁸⁴⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - Calcium repletion
 - IV fluid resuscitation ≤ 2d ⁽⁸⁵⁾
 - Diuretic, continuous
 - Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - Glucose 50%(0.50) with insulin ⁽⁷¹⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽¹¹³⁾

- **HOME, Both:**
 - Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽¹¹⁴⁾
 - Patient or caregiver demonstrates ability to manage care
 - Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽¹¹⁵⁾
 - Medication reconciliation ⁽¹¹⁶⁾
 - Follow-up care planned ⁽¹¹⁷⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽¹¹⁴⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(118, 119)
 - Skilled services, ≥ **One:** ⁽¹²⁰⁾
 - Skilled nursing ⁽¹²¹⁾

- Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽¹²²⁾
- Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽¹¹⁵⁾
 - Medication reconciliation ⁽¹¹⁶⁾
 - Follow-up care planned ⁽¹¹⁷⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽¹²³⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽¹²⁴⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽¹²¹⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(125, 126)
 - Functional impairments requiring at least supervision ⁽¹²⁷⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹¹⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽¹²¹⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(125, 126)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹²⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹¹⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(125, 126)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹²⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines

- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹¹⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(125, 126)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹²⁷⁾
 - Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽¹²⁸⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹¹⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽¹²⁹⁾
 - Ventilator dependent $\geq 6h/d$ and weaning planned ^(130, 131)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹³²⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽¹¹⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
054 NERVOUS SYSTEM NEOPLASMS WITH MCC	n/a	4	CMS GMLOS
055 NERVOUS SYSTEM NEOPLASMS WITHOUT MCC	n/a	3	CMS GMLOS
124 OTHER DISORDERS OF THE EYE WITH MCC OR THROMBOLYTIC AGENT	n/a	3.4	CMS GMLOS
125 OTHER DISORDERS OF THE EYE WITHOUT MCC	n/a	2.3	CMS GMLOS
146 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITH MCC	n/a	5.5	CMS GMLOS
147 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITH CC	n/a	3.6	CMS GMLOS
148 EAR, NOSE, MOUTH AND THROAT MALIGNANCY WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
180 RESPIRATORY NEOPLASMS WITH MCC	n/a	4.9	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
181 RESPIRATORY NEOPLASMS WITH CC	n/a	3.2	CMS GMLOS
182 RESPIRATORY NEOPLASMS WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
314 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH MCC	n/a	5.1	CMS GMLOS
315 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH CC	n/a	2.7	CMS GMLOS
316 OTHER CIRCULATORY SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	1.8	CMS GMLOS
374 DIGESTIVE MALIGNANCY WITH MCC	n/a	5.6	CMS GMLOS
375 DIGESTIVE MALIGNANCY WITH CC	n/a	3.6	CMS GMLOS
376 DIGESTIVE MALIGNANCY WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
435 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITH MCC	n/a	4.8	CMS GMLOS
436 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITH CC	n/a	3.4	CMS GMLOS
437 MALIGNANCY OF HEPATOBILIARY SYSTEM OR PANCREAS WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
542 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITH MCC	n/a	5.2	CMS GMLOS
543 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITH CC	n/a	3.5	CMS GMLOS
544 PATHOLOGICAL FRACTURES AND MUSCULOSKELETAL AND CONNECTIVE TISSUE MALIGNANCY WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
595 MAJOR SKIN DISORDERS WITH MCC	n/a	5.7	CMS GMLOS
596 MAJOR SKIN DISORDERS WITHOUT MCC	n/a	3.5	CMS GMLOS
597 MALIGNANT BREAST DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
598 MALIGNANT BREAST DISORDERS WITH CC	n/a	3.4	CMS GMLOS
599 MALIGNANT BREAST DISORDERS WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
606 MINOR SKIN DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
607 MINOR SKIN DISORDERS WITHOUT MCC	n/a	3	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
643 ENDOCRINE DISORDERS WITH MCC	n/a	5.1	CMS GMLOS
644 ENDOCRINE DISORDERS WITH CC	n/a	3.6	CMS GMLOS
645 ENDOCRINE DISORDERS WITHOUT CC/MCC	n/a	2.7	CMS GMLOS
682 RENAL FAILURE WITH MCC	n/a	4.4	CMS GMLOS
683 RENAL FAILURE WITH CC	n/a	3.1	CMS GMLOS
684 RENAL FAILURE WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
686 KIDNEY AND URINARY TRACT NEOPLASMS WITH MCC	n/a	5.3	CMS GMLOS
687 KIDNEY AND URINARY TRACT NEOPLASMS WITH CC	n/a	3.3	CMS GMLOS
688 KIDNEY AND URINARY TRACT NEOPLASMS WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
722 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5.5	CMS GMLOS
723 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.5	CMS GMLOS
724 MALIGNANCY, MALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2	CMS GMLOS
754 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5.1	CMS GMLOS
755 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.3	CMS GMLOS
756 MALIGNANCY, FEMALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
BRAIN MALIGNANCY OR METASTASIS	6%	5.5	InterQual
MALIGNANT DISEASE	18%	5.2	InterQual
SPINAL CORD COMPRESSION	16%	6.1	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

3:

Tachycardia and orthostatic hypotension are common symptoms of dehydration when vomiting or diarrhea are present (Tomas Nguyen and Saadia Akhtar, In: Rosen's Emergency Medicine, 9 edn. 2018:1129-49.e2).

4:

Orthostatic hypotension is a common complication for patients after chemotherapy or immunotherapy due to lack of oral intake and progressive dehydration. These patients frequently require close monitoring and intravenous fluids (Vecchie et al., Cardiooncology 2021, 7: 40).

5:

Repeated intravenous fluid boluses may be used to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as, lactated ringers (Myburgh, N Engl J Med 2018, 378: 862-3).

In adults, bolus volumes may be generally defined as fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34).

For patients with decreased cardiac reserve, significant renal dysfunction or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206).

6:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, emergency department (prior to decision to admit), or hospital Observation setting (if appropriate for this condition). These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

7:

Dexamethasone is effective in relieving chemotherapy-induced nausea and vomiting. Patients receiving chemotherapy with a high or moderate emetic risk may be given a combination of dexamethasone, serotonin antagonists, and/or aprepitant (Olver et al., Support Care Cancer 2017, 25: 297-301).

8:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

9:

The routes for outpatient analgesics include:

- Oral (PO)
- Subcutaneous (SC)
- Intranasal
- Transdermal

10:

Pain management is an essential component and effective therapy for patients with cancer (Li et al., Cochrane Database Syst Rev 2021, 8: CD011108; Swarm et al., J Natl Compr Canc Netw 2019, 17: 977-1007; WHO Guidelines for the Pharmacological and Radiotherapeutic Management of Cancer Pain in Adults and Adolescents. 2018).

11:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

12:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138; Wang et al., *Am J Nephrol* 2012, 35: 349-55).

13:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

14:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

15:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., *Crit Care Med* 2021, 49: e1063-e143; Rhodes et al., *Crit Care Med* 2017, 45: 486-552; Winters et al., *J Emerg Med* 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, *J Crit Care* 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

16:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; Pistolesi et al., *J Nephrol* 2018, 31: 797-812; Joannidis et al., *Intensive Care Med* 2017, 43: 730-49; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

17:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

18:

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii.-138).

19:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute

renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

20:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

21:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

22:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

23:

Paresis refers to a condition of muscle weakness with a reduction in voluntary movement while paralysis refers to a complete loss of movement. These criteria include, but are not limited to, paresis or paralysis of the face as well as difficulty swallowing (i.e., dysphagia).

Instruction: Patients experiencing general weakness or inability to move limbs from a non-neurologic cause are excluded from these criteria.

24:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

25:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

26:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications,

allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion
- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

27:

Graft versus Host Disease (GVHD) is a transplant complication that occurs when immunocompetent cells from a non-identical donor are grafted into an immunosuppressed recipient. The immunocompetent donor/graft cells recognize alloantigens in the host cells and the host develops a Graft vs Host (GVH) reaction. Classic acute GVHD typically presents during the first 100 days after a hematopoietic stem cell transplantation (HSCT). Classic chronic GVHD presents beyond the first 100 days post HSCT and may include overlap syndrome, with acute GVHD features. Classic acute GVHD affects the skin, gut, and liver. Diagnosis is strongly supported if symptoms include exanthema (an eruptive skin rash oftentimes associated with a fever or systemic symptoms), diarrhea, and elevated bilirubin levels. Chronic GVHD, in addition to skin, liver, or gut involvement, may also affect the mouth, eyes, respiratory tract, musculoskeletal system, and peripheral nerves. It presents with symptomatology resembling autoimmune diseases (e.g., systemic lupus, erythematosus, or rheumatoid arthritis).

Chronic GVHD grade can be categorized as mild, moderate, or severe. In patients with GVHD, hyperbilirubinemia and small intestinal/colonic involvement are known risk factors for increased mortality (Hill et al., Annu Rev Immunol 2021, 39: 19-49; Penack et al., Lancet Haematol 2020, 7: e157-e67; Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50).

In determining GVHD grade, stages for skin, liver and gut are scored.

Acute GVHD Staging and Grading (Rodrigues et al., American journal of clinical dermatology 2018, 19: 33-50):

Grade I: Stage 1-2 skin organ involvement, only

Grade II: Stage 3 skin; Stage 1 liver and gut

Grade III: Stage 1-3 skin; Stage 2-3 liver and Stage 2-4 gut

Grade IV: Stage 4 skin and liver

Stage 1

- Skin: Rash < 25%(0.25) of body surface area
- Liver: Bilirubin 2-3 mg/dL (35-40 µmol/L)
- Gut: Diarrhea > 500-1000 ml/day

Stage 2

- Skin: Rash 25%(0.25) to 50%(0.50) of body surface area with associated symptoms
- Liver: Bilirubin 3-6 mg/dL (53-102 µmol/L)
- Gut: Diarrhea > 1000 ml/day

Stage 3

- Skin: Generalized rash covering > 50%(0.50) of body surface area
- Liver: Bilirubin 6-15 mg/dL (104-225 µmol/L)
- Gut: Diarrhea > 1500 ml/day

Stage 4

- Skin: Generalized erythroderma (reddening of skin often associated with exfoliation) or bullous eruption

- Liver: Bilirubin > 15 mg/dL (225 µmol/L)
- Gut: Severe abdominal pain with or without ileus

Chronic GVHD grading (scores are based on a 4-point scale of 0-3):

- Mild: 1 or 2 organs, with score no more than 1 with lung score 0
- Moderate: 3 or more organs with no more than score 1 or > 1 organ (not lung) with score 2 or lung score 1
- Severe: At least 1 organ with score of 3 or lung score of 2 or 3

28:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

29:

Medication adjustments tend to be driven by patient response. Patients being treated for symptoms of rejection usually have a biopsy and their medications adjusted based on the biopsy results to control the rejection.

30:

All medications commonly used to prevent opportunistic infections or support patients pre and post-transplant cause some degree of side effects. One of the goals of treatment is to titrate the dosage to minimize toxicity while maximizing benefit. Often this fine-tuning is empiric. If a patient has an intolerable side effect from a particular drug, an alternative medication or different combination or dosage of medications is tried.

31:

Hyperviscosity syndrome is caused by elevated immunoglobulin levels which coat red and white blood cells, causing increased blood viscosity, sludging of blood and hypoperfusion. It can be seen in diseases such as multiple myeloma, leukemia, polycythemia, and other myeloproliferative disorders. Symptoms of hyperviscosity syndrome include headache, tinnitus, stupor, coma, and uncontrolled or recurrent epistaxis. In the presence of symptoms, therapeutic plasma exchange is indicated (Halaseh et al., QJM 2022, 115: 30-1; Emerson and Kaide, Emerg Med Clin North Am 2018, 36: 603-8; Gertz, Blood 2018, 132: 1379-85).

32:

Apheresis is a procedure that separates and removes elements in the blood (e.g., white blood cells (WBCs), platelets, stem cells, plasma bound organic compounds, or antibodies). Subsequent replacement with blood components, solvent detergent plasma (SDP), albumin, and/or normal saline may follow apheresis.

33:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

34:

Diarrhea is one of the most common adverse effects of cancer and cancer treatment and is a frequent symptom of colitis. Diarrhea is often made worse by preexisting inflammatory bowel disease. Per the National Comprehensive Cancer Network (NCCN) Guidelines, patients experiencing diarrhea related to cancer treatments should be treated with steroids, immunosuppressants, and/or monoclonal antibody therapies (Ajani et al., J Natl Compr Canc Netw

2022, 20: 167-92; Bellaguarda and Hanauer, Am J Gastroenterol 2020, 115: 202-10; Gong and Wang, JCO Oncol Pract 2020, 16: 453-61; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

35:

Abdominal computed tomography (CT) is generally performed when patients with severe diarrhea present with abdominal pain to diagnose colitis or to rule out other intra-abdominal processes (Tang et al., Front Immunol 2021, 12: 800879; Gong and Wang, JCO Oncol Pract 2020, 16: 453-61; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

36:

Corticosteroids are the first line treatment for most inhibitor-related toxicities secondary to immunotherapy (Ben-Horin et al., Clin Gastroenterol Hepatol 2022, 20: 2868-75 e1; Fan and Naidoo, JCO Oncol Pract 2020, 16: 462-3; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

37:

Immunosuppressant therapies to treat diarrhea related to cancer or cancer treatments can include but are not limited to (Fan and Naidoo, JCO Oncol Pract 2020, 16: 462-3; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913):

- Infliximab
- Tofacitinib
- Ustekinumab
- Vedolizumab

38:

Intractable pain can include but is not limited to pain that is:

- Unresponsive to ≥ 3 doses of an analgesic (includes PO) within 24 hours
- Unresponsive to a continuous analgesic within last 24 hours
- Unresponsive to a transdermal analgesic ≥ 24 hours

39:

Jaundice can be a sign of advanced disease or another complication and therefore requires a thorough workup at the Acute level of care (Kastelijns et al., J Gastrointest Cancer 2022;; An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Lee et al., HPB (Oxford) 2018, 20: 477-86).

40:

Malignant pleural effusions are most frequently found in patients with lung cancer, breast cancer, and lymphoma. A malignant pleural effusion is a buildup of fluid in the pleural space in which cancerous cells are present in the fluid. Treatment for malignant pleural effusions differs based on goals of therapy, patient functional status, and life expectancy (Bibby et al., Eur J Cardiothorac Surg 2019, 55: 116-32; Feller-Kopman et al., Am J Respir Crit Care Med 2018, 198: 839-49).

41:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

42:

In patients with malignant pleural effusions, chest tube insertion includes pleural drains and indwelling pleural catheters. Chest tubes are frequently placed as a palliative measure when patients with malignant pleural effusions are having recurrent pleural effusions. Chest tubes are also placed in anticipation of the introduction of a sclerosing agent (Bibby et al., Eur J Cardiothorac Surg 2019, 55: 116-32; Feller-Kopman et al., Am J Respir Crit Care Med 2018, 198: 839-49; Reddy et al., Ann Am Thorac Soc 2019, 16: 17-21).

43:

Small bore chest tubes are less than or equal to 14F and large bore chest tubes are greater than 14F (Mummadi et al., Ann Emerg Med 2020, 76: 88-102; Chang et al., Chest 2018, 153: 1201-12).

44:

Chest tube insertion is an appropriate intervention for patients with malignant pleural effusions. Chest tubes, also frequently called indwelling pleural catheters (IPCs), can be placed solely for fluid drainage or they can be placed as a means to administer sclerosing agents. If the administration of a sclerosing agent fails, an IPC may be left in place to continuously drain fluid from the pleural space (Reddy et al., Ann Am Thorac Soc 2019, 16: 17-21).

45:

Sclerosing agents are administered into the pleural space in an attempt to form adhesions and prevent the reaccumulation of fluid. There are various effective sclerosing agents available and the selection of agent is based on the needs of the patient (Dipper et al., Cochrane Database Syst Rev 2020, 4: Cd010529; Feller-Kopman et al., Am J Respir Crit Care Med 2018, 198: 839-49).

46:

Recommendations regarding evaluation, treatment, and setting of care for patients with fever and neutropenia are based on risk stratification into low and high risk groups. Assessment of the risk for complications from a severe infection should be performed upon presentation.

Low risk patients are generally defined by clinical stability without an active medical comorbidity (e.g., mucositis, nausea and vomiting, mental status change, etc.). High risk factors may include, but are not limited to, the following:

- Absolute neutrophil count (ANC) $\leq 100/\text{cu.mm}(100 \times 10^6/\text{L})$
- Mental status change
- Chronic lung disease
- Abdominal pain, new onset
- Mucositis
- Nausea and vomiting
- Diarrhea
- Uncontrolled pain
- Underlying malignancy with bone marrow involvement (e.g., acute leukemia)

In addition, use of validated risk assessment tools which take into account a multitude of risk factors, such as the Multinational Association for Supportive Care in Cancer (MASCC) scoring system, are recommended to categorize patients into low and high risk groups. Pediatric guidelines specifically recommend that hospitals adopt and use a validated risk stratification tool to guide routine management. Select low risk patients may be candidates for outpatient management, but inpatient management remains the standard of care (Pettit et al., Am J Emerg Med 2021, 50: 5-9; Zimmer and Freifeld, J Oncol Pract 2019, 15: 19-24).

47:

Febrile neutropenia is a medical emergency for oncology and hematology patients. The Multinational Association for Supportive Care in Cancer (MASCC) Index is the recommended tool for the risk stratification of these patients (Pettit et al., Am J Emerg Med 2021, 50: 5-9).

48:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than $500/\text{cu.mm}(500 \times 10^6/\text{L})$ are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm}(1.5 \times 10^9/L)$

$= 330/\text{cu.mm}(330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

49:

An absolute neutrophil count that is expected decline to $< 500/\text{cu.mm}(500 \times 10^6/L)$ within 48 hours is included in the definition of neutropenia (National Comprehensive Cancer Network, The NCCN clinical practice guidelines in oncology, Prevention and Treatment of Cancer-Related Infections (Version 2.2023). 2023).

50:

Oral mucositis is a common side effect of cancer treatments. More than one-third of cancer patients experience mucositis whether it be moderate or severe during the course of their cancer treatment. Mucositis, if left untreated, can lead to anorexia, weight loss, malnutrition and dehydration (Hong et al., Support Care Cancer 2019, 27: 3949-67; Baden et al., J Natl Compr Canc Netw 2016, 14: 882-913).

51:

Instruction: For patients with spinal cord compression requiring radiation or chemotherapy refer to the Hematology/Oncology: Treatments subset.

52:

Spinal metastases is a common complication of cancer and considered an oncological emergency not only due to back pain, but more importantly, can exhibit progressive motor and sensory loss and lead to permanent damage. Steroids, radiotherapy, and surgery are the primary treatments for spinal cord compression caused by spinal metastases (Van den Brande et al., J Bone Oncol 2022, 35: 100446; Kumar et al., Clin Spine Surg 2017, 30: 156-63; George et al., Cochrane Database Syst Rev 2015, 2015: CD006716).

53:

Instruction: For known Von Willebrand disease, use the Anemia/Bleeding subset.

54:

Thrombocytopenia is a frequent complication of cancer and cancer treatments. The most common treatments for thrombocytopenia are blood product transfusions, corticosteroids and immunoglobulins (Omar et al., Front Immunol 2020, 11: 1354; Schiffer et al., J Clin Oncol 2018, 36: 283-99).

55:

Intravenous immunoglobulin (IVIG) is a blood product which contains immunoglobulins extracted, then pooled, from the plasma of thousands of blood donors. Its administration carries the same risks as those of any blood product transfusion, including headache, backache, chills, nausea, muscle pain, and, although rare, anaphylaxis. IVIG is used for a wide range of disorders, and its therapeutic mechanism varies based on the dose given and the patient's disease. IVIG is used for the treatment of diseases in neurology, hematology, immunology, dermatology, and nephrology.

56:

Tumor lysis syndrome can be a serious complication of cancer treatments. It is characterized by the release of intracellular contents into the blood vessels causing metabolic abnormalities. If untreated, it can lead to renal failure, arrhythmias, seizures, loss of muscle control, or death (Alqurashi et al., Cureus 2022, 14: e30652; Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

57:

Aggressive hydration is an essential component of prevention and treatment of tumor lysis syndrome (Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

58:

It is recommended that tumor lysis syndrome prophylaxis be initiated prior to chemotherapy. However, rasburicase is considered the first line agent in the treatment of tumor lysis syndrome. Allopurinol is an alternative therapy when rasburicase is contraindicated due to hypersensitivity reaction or G6PD deficiency (Zelenetz et al., Journal of the National Comprehensive Cancer Network 2021, 19: 1218-30).

59:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

60:

Grades 1 and 2 of cytokine release syndrome (CRS) are usually characterized by a fever alone or fever with hypoxia and/or hypotension, respectively. Interventions for CRS can include broad spectrum corticosteroids, IV fluids, and anticytokine therapies. If fever persists, tocilizumab which is usually given to Grade 2 CRS can be considered (Thompson et al., J Natl Compr Canc Netw 2022, 20: 387-405; Santomaso et al., J Clin Oncol 2021, 39: 3978-92; Lee et al., Biol Blood Marrow Transplant 2019, 25: 625-38).

61:

Cytokine release syndrome can occur after immunotherapies such as CAR (chimeric antigen receptor) T-cell therapy and it can also occur after bone marrow or stem cell transplants. It is characterized by a fever and a systematic inflammatory response (Thompson et al., J Natl Compr Canc Netw 2022, 20: 387-405; Abboud et al., Bone Marrow Transplant 2021, 56: 2763-70; Santomaso et al., J Clin Oncol 2021, 39: 3978-92).

62:

Instruction: In some institutions, the first 3 doses of muromonab-CD3 are required to be given in Intermediate care when the patient is experiencing acute rejection. The initial doses of muromonab-CD3 can cause violent headaches, fevers, and dyspnea. Dyspnea is more common in fluid overloaded patients and can resemble acute respiratory distress syndrome (ARDS) in severe forms.

63:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in

treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

64:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

65:

Aphasia for patients with neurotoxicity can include but is not limited to episodes of expressive, global, or receptive aphasia.

66:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

67:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

68:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

69:

Hyperkalemia (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50; Depret et al., Ann Intensive Care 2019, 9: 32; Montford and Linas, J Am Soc Nephrol 2017, 28: 3155-65).

70:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

71:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

72:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

73:

Refractory seizures, also known as uncontrolled, intractable, or drug-resistant seizures, are resistant to treatment and are poorly controlled with medication. Recurrent seizures are seizures that occur more than once and may occur without warning (Billington et al., J Clin Med 2016, 5:).

74:

Refractory arrhythmias, also known as uncontrolled, intractable, or drug-resistant arrhythmias, are resistant to treatment and are poorly controlled with medication. Recurrent arrhythmias are arrhythmias that occur more than once and may occur without warning.

75:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15

minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

76:

Blood pressure is considered labile when it is extremely variable and difficult to control.

77:

Hyperosmolar therapy creates an osmotic gradient, facilitating the removal of water across the blood brain barrier, resulting in a decreased brain volume and a decrease in intracranial pressure. While hypertonic saline and mannitol are the 2 most used hyperosmolar agents, there is insufficient evidence to recommend one over the other for improving neurological outcomes in patients with acute ischemic stroke (Cook et al., Neurocrit Care 2020, 32: 647-66). Doses and strengths are generally determined by institutional practice patterns. Common side effects associated with osmotic therapy include dehydration, hypotension, renal failure, and acute respiratory distress.

78:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

79:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

80:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-

50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

81:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

82:

Abnormalities of calcium can have significant impact on morbidity and mortality due to its crucial role in neuromuscular and cardiac function. Patients who present with symptoms usually have an abrupt decline in ionized calcium.

Common symptoms of hypocalcemia include:

- Perioral numbness
- Carpopedal spasm of the hands or feet
- Flaccid paralysis
- Muscle weakness
- Paresthesia
- Tetany

Symptomatic hypocalcemia requires treatment with intravenous calcium (Turner et al., Endocr Connect 2019, 8: X1).

83:

Calcium levels within criteria represent total serum calcium.

84:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

85:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zeng et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

86:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

87:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

88:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

89:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

90:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly

established baseline, or parameters that the medical practitioner determines are acceptable.

91:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

92:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

93:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

94:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

95:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

96:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

97:

Graft dysfunction is usually treated empirically with anti-rejection therapy while awaiting the results of the biopsy. Immunosuppressive medications are initiated based on the signs and symptoms of organ dysfunction or graft versus host disease (GVHD).

98:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

99:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

100:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

101:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

102:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

103:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

104:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

105:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

106:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

107:

Neurological deficit(s) may include:

- Altered level of consciousness
- Cognitive deficit
- Motor deficit
- Speech deficit
- Sensory deficit

Neurological disorder may include:

- Amyotrophic lateral sclerosis (ALS)
- Dementia (including Alzheimer's disease)
- Guillain-Barre syndrome (GBS)
- Multiple sclerosis (MS)
- Myasthenia gravis (MG)
- Parkinson's disease (PD)
- Transverse myelitis

108:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

109:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

110:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

111:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

112:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

113:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

114:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

115:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

116:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

117:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50%

(0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

118:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

119:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

120:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

121:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

122:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

123:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

124:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

125:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

126:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening

- Wheelchair mobility
- Ambulation

127:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

128:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

129:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

130:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

131:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder

- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

132:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.